

Converting between Sequent Systems and Truth Trees

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April 17, 2026

Sequent System Rules (Propositional Calculus)

The sequent system we are using has the following rules of operation. Γ and Δ are sets (multisets) and A and B are atomics.

Operational Rules

Negation

$$\frac{\Gamma \Rightarrow \Delta, A}{\neg A, \Gamma \Rightarrow \Delta} \neg L \qquad \frac{A, \Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, \neg A} \neg R$$

Conjunction

$$\frac{A, B, \Gamma \Rightarrow \Delta}{A \wedge B, \Gamma \Rightarrow \Delta} \wedge L \qquad \frac{\Gamma \Rightarrow \Delta, A \quad \Gamma \Rightarrow \Delta, B}{\Gamma \Rightarrow \Delta, A \wedge B} \wedge R$$

Disjunction

$$\frac{A, \Gamma \Rightarrow \Delta \quad B, \Gamma \Rightarrow \Delta}{A \vee B, \Gamma \Rightarrow \Delta} \vee L \qquad \frac{\Gamma \Rightarrow \Delta, A, B}{\Gamma \Rightarrow \Delta, A \vee B} \vee R$$

Implication

$$\frac{\Gamma \Rightarrow \Delta, A \quad B, \Pi \Rightarrow \Lambda}{A \rightarrow B, \Gamma, \Pi \Rightarrow \Delta, \Lambda} \rightarrow L \qquad \frac{A, \Gamma \Rightarrow \Delta, B}{\Gamma \Rightarrow \Delta, A \rightarrow B} \rightarrow R$$

Structural Rules

Weakening

$$\frac{\Gamma \Rightarrow \Delta}{A, \Gamma \Rightarrow \Delta} WL \qquad \frac{\Gamma \Rightarrow \Delta}{\Gamma \Rightarrow \Delta, A} WR$$

Contraction

$$\frac{A, A, \Gamma \Rightarrow \Delta}{A, \Gamma \Rightarrow \Delta} CL \qquad \frac{\Gamma \Rightarrow \Delta, A, A}{\Gamma \Rightarrow \Delta, A} CR$$

Exchange

$$\frac{\Gamma, A, B, \Pi \Rightarrow \Delta}{\Gamma, B, A, \Pi \Rightarrow \Delta} XL \qquad \frac{\Gamma \Rightarrow \Delta, A, B, \Lambda}{\Gamma \Rightarrow \Delta, B, A, \Lambda} XR$$

File Layout

The sequent system will be formatted as a json with a list of nodes containing their node ID, antecedent/succedent, the node ids used to get there and the rule used to get there, going from goal upwards. For example, the following sequent:

$$\frac{\frac{\frac{A \Rightarrow A}{B, A \Rightarrow A} WL}{A, B \Rightarrow A} XL \quad \frac{\frac{B \Rightarrow B}{A, B \Rightarrow B} WL}{A, B \Rightarrow A \wedge B} \wedge R}{A, B \Rightarrow A \wedge B} \wedge R$$

would be represented as: !!! TODO !!!

Truth Tree Rules (Propositional Calculus)

Decomposition Rules

Closure Rules

File Layout

The truth tree will be formatted into a json-like form for use in Willow.

Putting them together